MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026
FOURTH SEMESTER
Value Addition Course (VAC) - MG4VACGEO200 - GEOLOGY AND ENVIRONMENT
(2024 ADMISSION ONWARDS)

QP CODE: 25901295

ANSWER KEY

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

PART A

(Short Answer in 60 words; Answer any 7 out of 8; 2 marks each)

1. Natural hazard management and mitigation, Waste management and remediation (Any two; 2 marks) [U] [CO1]
2. Natural resources- substances or materials found in nature that can be exploited for economic gain or to sustain life. (Definition; 2 marks) [K] [CO3].
3. Habitat destruction and land degradation, leading to the loss of biodiversity and the displacement of local flora and fauna. (2 marks) [K] [CO5].
4. Biotic and Abiotic components (2 marks) [K] [CO2]
5. Rainwater harvesting-collection and storage of rainwater from surfaces like rooftops or open ground for future use, rather than allowing it to run off. (2 marks) [U] [CO4]
6. Landslide is the downward movement of a mass of rock, debris, or earth under the direct influence of gravity. (Definition; 2 marks) [K] [CO5]
7. Thermosphere is the layer of the Earth's atmosphere located directly above the mesosphere, characterized by temperatures that increase significantly with altitude due to the absorption of high-energy solar radiation. (Definition; 2 marks) [K] [CO2]
8. Algal bloom is a rapid increase or accumulation in the population of algae in freshwater or marine water systems, often resulting in a visible green, red, or brown discoloration of the water surface (Definition; 2 marks). [U] [CO4]

Part B

Write short notes in 250 words; Answer any 3 questions

Each question carries 8 marks

9. Environmental hotspot (Definition-2 marks; Explanation-2 marks); Environmental hotspots in India (4 marks) [U] [CO2]
10. Natural resources (Definition-2 marks); importance of preservation of natural resources (6 marks) [U] [CO3]

11. Floods (definition and explanation-3 marks); effect of floods on water quality in rivers (5 marks) [U] [CO5]
12. Groundwater pollution (definition-2 marks; health hazards associated with groundwater pollution (6 marks). [U] [CO4]

13. Ozone layer (formation-2 marks; depletion- 2 marks; effects and prevention of ozone depletion (4 marks) [U] [CO4]

Part C

Write short notes in 400 words; Answer any 1 question

Each question carries 12 marks

14. Air pollution (Definition- 2 marks); Water pollution (Definition- 2 marks); environmental pollution (Definition- 2 marks); explanation on interconnection between air and water pollution (6 marks) [E] [CO4]
15. EIA (Definition- 2 marks); Stages of EIA with sequence and purpose (10 marks). [U] [CO5]

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